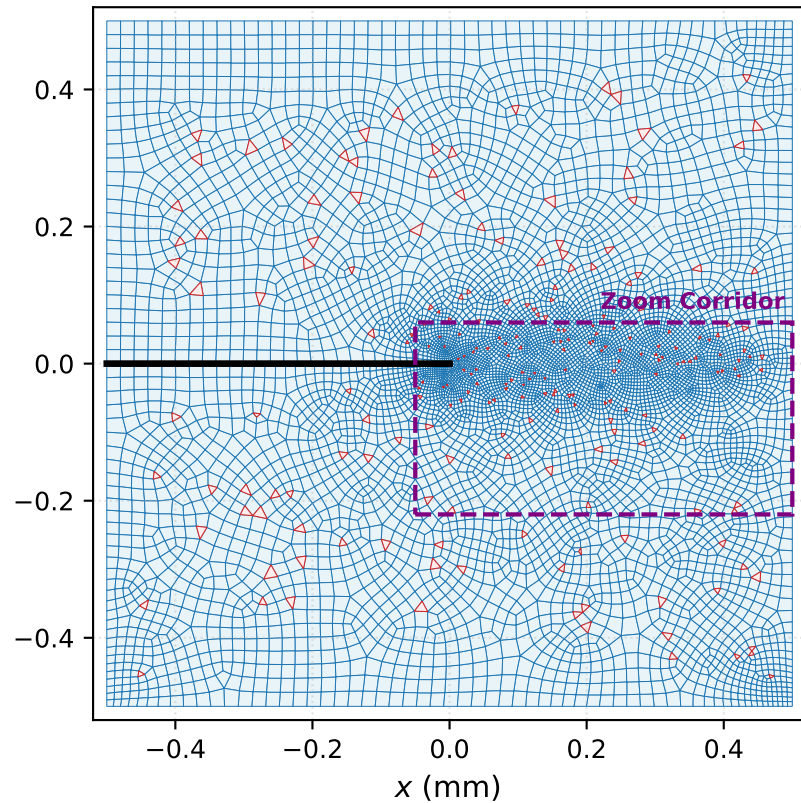


Figure 4: Final Abaqus-Native Adaptive Mesh Topology (errorTarget=5.0%, 7,865 Elements)

(a) Full Adaptive Domain (7,865 Finite Elements)



(b) Crack Zone Detail ($h_{\min} = 2.1 \mu\text{m}$, Hybrid Quads/Tris)

